Lecture 3 - Recap & Copilot Strategy

Management Science

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Working with AI Assistants

Context Engineering

Write clear descriptions of what you want:

```
# Create subplot with 2 rows, 1 column
# Top: line plot of revenue over time
# Bottom: bar chart of profit by quarter
```

. . .

Tip

- **Be specific** in your instructions
- **Review** generated code
- **Test the code** as it might use old syntax
- **Iterate** for better results
- **Force To-Tos** so everything is completed

Watch Out For

- **Deprecated methods**: `plt.subplot()` vs `plt.subplots()`
- **Missing imports**: Always verify imports are included
- **Wrong assumptions**: Might guess your data structure incorrectly
- **Over-complexity**: Very often suggests unnecessary features

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Remember: Generative AI is stochastic!

What We Learned

Foundation Complete!

Why This Matters?

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You now have a good foundation for data-driven decisions!

What's Next

Preview: 4 - Monte Carlo

Next Session: Modeling Business Uncertainty

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We'll combine everything you've learned:

- NumPy for random number generation
- Visualization for showing probability distributions

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Real applications:

- Predict project completion times
- Estimate financial risks
- Make decisions under uncertainty

The End

That's it for today! **Make sure you:**

- ☐ Have completed all the notebooks
- ☐ Check whether you could follow so far
- ☐ Set up Copilot

. . .

> Tip
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> Every line of code you write makes you a better programmer. Every concept you understand makes you a better decision-maker. Keep practicing and keep learning!

Literature

Interesting Literature on Algorithms

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[1]
- Ferguson, T.S. (1989) 'Who solved the secretary problem?', Statistical Science, 4(3). doi:10.1214/ss/1177012493.

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Here

[1] A great inspiration to learn more about Algorithms!

- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

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> **ℹ Note**
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> Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

More Literature

For more interesting literature, take a look at the literature list of this course.

Bibliography